INCH-POUND

MIL-L-87981A

26 October 1994

SUPERSEDING
MIL-L-87981
15 Apr 88

MILITARY SPECIFICATION

LOADER, BACKHOE, SCOOP-TYPE, PNEUMATIC TIRED

1. SCOPE

1.1 Scope. This purchase description covers a commercial , diesel powered, four wheel, air transportable, scoop-type backhoe loader.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the current issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

VV-F-800	Fuel Oil, Diesel

MILITARY

MIL-T-3351	Tractor, Full Tracked, Low Speed, Tractor Wheeled, Agricultural and Tractor Wheeled, Industrial and their Attachments. Packaging of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/LVRE, 225 Ocmulgee Ct, Robins AFB, GA 31098-1647 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3805

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MILITARY (cont.)

MIL-T-83133	Turbine Fuel, Aviation, Kerosene Type, Nato F-34, Grade JP-8

STANDARDS

FEDERAL

FED-STD-595	Colors
FED-STD-297	Rustproofing of Commercial (Non-tactical) Vehicles

MILITARY

MIL-STD-129	Marking for Shipping and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-209	Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment
MIL-STD-889	Dissimilar Metals
MIL-STD-1474	Noise Limits for Military Material (Metric)
MIL-STD-1791	Designing For Internal Aerial Delivery in Fixed Wing Aircraft

HANDBOOKS

Air Force Systems Command DH-1-11 Air Transportability

2.1.2 Nongovernment standards and other publications.

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE) SAE Handbook

SAE J326	Nomenclature-Hydraulic Backhoes
SAE J517	Hydraulic hose
SAE J524	Seamless low carbon steel tubing annealed for bending and flaring.
SAE J525	Welded and cold drawn low carbon steel tubing annealed for bending and flaring.
SAE J534	Lubrication Fittings
SAE J1349	Engine Power Test Code-Spark Ignition and Diesel

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(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, Pa 15096).

THE TIRE AND RIM ASSOCIATION, INC. Year book

(Application for copies should be addressed to the tire and rim association, Inc., 3200 W. Market Street, Akron OH 44313).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

LAW AND REGULATIONS

29CFR 1926.600	Equipment
29CFR 1926.602	Material Handling Equipment
29CFR 1926.1000	Rollover Protective Structures (ROPS) for Material Handling Equipment
29CFR 1926.1003	Overhead Protection for operators of Agricultural and Industrial Tractors

(The code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription by the superintendent of documents, US Government Printing Office, Washington DC 20402. When indicated, reprints of certain regulations may be obtained from the federal agency responsible for issuance thereof).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associate detail specifications and specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article test sample. Unless otherwise specified (see 6.2), the contractor shall furnish a complete unit for testing and inspection in accordance with sections 4 and 6. The first article test sample may be a preproduction item or a production item which meets the requirements of this specification. The sample shall be identical to the items procured under terms of the contract. Approval of the first article sample shall not constitute a waiver by the government of its rights under the provisions of the contract.

3.2 Design. The loader-backhoe shall be hydraulically operated, diesel engine driven, pneumatic-tired and shall have two front steerable wheels and two rear drive wheels. The unit shall have a full power shift transmission and be air transportable. All components and the unit as a whole shall be

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designed to operate effectively under extreme conditions. The loader-backhoe shall have as a minimum 90 net horsepower in accordance with SAE test procedure J1349.

3.2.1 Sealed Cab. When specified (see 6.2), a fully enclosed lockable type ROPS cab shall be provided. The cab shall be the manufacture's current standard production type enclosing the operator's seat and all controls and shall be equipped with front and rear power-operated windshield wipers. When specified (see 6.2), a personnel heater, defroster and/or defroster fan, or air-conditioners or combination thereof shall be provided. The cab shall provide doors or emergency exit provisions on each side.

3.2.2 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.2.3 Prohibited Materials.

3.2.3.1 ELIMINATION OF USE OF CLASS 1 OZONE DEPLETING SUBSTANCES

3.2.3.1.1 In accordance with Section 326 of the National Defense Authorization Act for Fiscal Year 1993 (Public Law 102-484) the Air Force may not award a contract after 1 Jun 93 (unless a waiver to do so has been obtained) which includes a specification, standard, drawing or other document that requires the use of a Class 1 ODS in the design, manufacture, test, operation, or maintenance of the system, subsystem, item, component or process, that establishes a requirement that can only be met by the use of a Class 1 ODS, or that requires the delivery of any item of supply that contains a Class 1 ODS or any service that includes the use of a Class 1 ODS.

3.2.3.1.2 The Contractor is requested to review the Government specifications, standards, drawings or other documents provided with or referenced in this solicitation and submit the following information with their offer to the Contracting Officer whose name appears on the front of this solicitation:

(a) A statement that the government specification, standard, drawing or other document requires/does not require (delete one) the use of a Class 1 ODS in the design, manufacture, test, operation or maintenance of the system, subsystem, item, component or process, and if so, identify;

(b) A statement that the government specification, standard, drawing or other document does/does not (delete one) establish a requirement that can only be met by the use of a Class 1 ODS, and if so, identify; and

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(c) A statement that this solicitation does/does not (delete one) require the delivery of any item of supply that contains a Class 1 ODS or any service that includes the use of a Class 1 ODS, and if so, identify.

3.2.3.1.3 For the purpose of this notice, a Class 1 ODS consists of any kind of the following:

CFC-11 ----- Trichlorofluoromethane
CFC 12 ----- Dichlorodifluoromethane
CFC 113 ---- Trichlorotrifluoroethane
CFC-114 ---- Dichlorotetrafluoroethane
CFC-115 ---- Chloropentafluoroethane

HALON 1011 - Bromochloromethane
HALON 1202 - Dibromodifluoromethane
HALON 1211 - Bromochlorodifluoromethane
HALON 1301 - Bromotrifluoromethane
HALON 2402 - Dibromotetrafluoroethane

CFC 13 -- Chlorotrifluoromethane
CFC 111 - Pentachlorofluoroethane
CFC 112 - Tetrachlorodifluoroethane
CFC 211 - Heptachlorofluoropropane
CFC 212 - Hexachlorodifluoropropane
CFC 213 - Pentachlorotrifluoropropane
CFC 214 - Tetrachlorotetrafluoropropane
CFC 215 - Trichloropentafluoropropane
CFC 216 - Dichlorohexafluoropropane
CFC 217 - Chloroheptafluoropropane

Carbon Tetrachloride - Tetrachloromethane

Methyl Chloroform ---- Trichloromethane

Methyl Bromide ------- Bromomethane; Monobromomethane

3.2.3.1.4 The use of wood and/or wood products is prohibited. Similarly, vinyl, polyvinylchloride, polyester, and corrosive type RTV (yields acetic acid upon curing) are prohibited for use in all forms, interior and exterior. The use of asbestos in any form is prohibited.

3.2.3.2 Protective treatment. Materials that are subject to deterioration shall be protected in such a manner that shall in no way prevent compliance with the performance requirements of this specification. Protective coatings that will chip, crack, or scale with age or extremes of climatic conditions shall not be used. Surface preparation, finishes and coatings shall be IAW standard commercial practice.

3.2.3.2.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3 Commerciality. The manufacturer shall be experienced in designing and building loader-backhoes and shall have sold them to the general public for at least one year prior to the opening date of the solicitation. Upon request of the contracting officer, bidders shall submit evidence of the commerciality of their machines in the form of catalogs, commercial brochures and data. Additionally, these bidders shall furnish names and addresses of nongovernment sources which were sold equipment at least one year prior to the opening date of the solicitation. The unit shall be the current commercially available unit differing only as necessary to meet the requirements of this specification.

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3.4 Integrity. Each unit furnished shall be complete and fully serviced such that it is capable of providing its intended service upon arrival at its final destination.

a. Exceptions:

(1) When otherwise specified in section 5.

(2) When to do so would violate Federal, State or other applicable regulations, or create a potentially hazardous or damaging condition.

(3) When artic anti-freeze and artic lubrication are required at the final destination they will be added by the buying agency at destination.

3.5 Safety. The loader-backhoe shall be fitted with safety devices or coverings for all parts that present safety hazards. When specified (see 6.2) loader shall comply with 29CFR 1926.600, 29CFR 1926.602, 29CFR 1926.1000 and 29CFR 1926.1003. Backhoe shall possess an audible backup warning alarm triggered by putting the shift lever in reverse. Unless otherwise specified herein, the unit shall conform to all Federal Laws and Regulations governing noise, air pollution and other health and safety requirements which are in effect for this type of equipment at the time of contract award.

3.5.1 Sound levels. The maximum sound level produced by the vehicle shall not exceed 85 dB(A) when tested in accordance with paragraph 4.4.9 below. An additional 2 dB(A) allowance shall be permitted for production units to provide for variation in test site, temperature gradients, test equipment, and inherent differences in nominally identical vehicles.

3.5.1.1 Exterior sound levels. The vehicle exterior sound level shall not exceed 84 dB(A) when measured IAW the criteria and parameters of paragraphs 5.3.1 thru 5.3.2 of MIL-STD-1474b; vehicle type shall be classified as Construction and Materials Handling Equipment for this test. If the exterior sound level exceeds 84 dB(A), a noise hazard caution sign(s) shall be posted IAW paragraph 4.3 of MIL-STD-1474b. An exterior sound level of greater than 88 dB(A) at any time during the test shall not be acceptable.

3.5.1.2 When cab is required, the cab interior sound level shall not exceed 84 dB(A) when measured in accordance with the criteria and parameters of paragraphs 5.1.2 thru 5.1.2.5 of MIL-STD-1474b. The vehicle shall be operated in accordance with paragraph 5.1.2.3.1.2 of MIL-STD-1474b. A cab interior sound level of greater than 84 dB(A) at any time during the test shall not be acceptable. The vehicle speed shall not be greater than 3 MPH in accordance with 5.1.2.4.1 of MIL-STD-1474b.

3.6 Fastening devices. All screws, pins, bolts and similar parts shall be installed with means for preventing loss of tightness. All such parts, when subjected to removal or adjustment, shall not be swaged, peened, staked or otherwise permanently deformed.

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3.7 Lubrication. All bearings (except for sealed-for-life type), mating gears and sliding parts shall be provided with means for lubrication. The reservoirs of splash type systems shall be fitted with an acceptable method for checking the oil level. All parts normally requiring grease for lubrication shall be provided with lubrication fittings in accordance with SAE J534. Where the use of high lubrication pressures may damage grease seals, a pressure release device shall be provided. Extended fittings shall be provided to lubricate parts or assemblies which are not readily accessible for direct lubrication, or which are likely to be overlooked because of their inaccessibility.

3.8 Cleaning, priming and painting. All portions of the vehicle normally painted shall be cleaned, treated, primed and finish coated in accordance with the contractors standard commercial practice. All paints and primers shall be lead and chromate free. The top coat color shall be dark green Color Number 24052 of FED-STD-595. All exterior trim (such as wheel covers, wheels, hub caps, running boards, metal radiator grills, bumpers, head light rings and other exterior trim) shall match the exterior color of the vehicle. The finished product shall show no evidence of uneven application, curtains, runs, sags, orange peel, lack of adhesion or other paint defects.

3.9 Temperature conditions. Without preheating from an external source, the engine shall start within 5 minutes and the engine shall operate under full load conditions within 15 minutes after engine start in any ambient temperature from -20°F to +115°F. The use of starting aids is permissible.

3.9.1 Winterization. When specified (see 6.2), the backhoe loaders shall be equipped with a winterization system to permit storage at ambient temperatures of -60°F. Heaters shall operate on 110 volt alternating current (AC) and shall be wired through a junction block to a single three pronged (male) weatherproof slave receptacle for receiving external power and grounding the vehicle. A three wire connecting cable, extending at least 25 feet beyond the front or rear of the vehicle when connected to the slave receptacle and of adequate line capacity to supply power for all heater units, simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end and a standard weatherproof three pronged (two power, plus one ground) male connector at the other end. Electrical apparatus shall conform to FMCSR 393.77(c)(7). Electrical insulation of connecting cable shall withstand normal operating stresses in low ambient air temperature without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation and without loose excess wire. Storage of the cable shall be provided on the vehicle frame. Heaters shall be furnished as follows:

a. Coolant heater, 1,500 watt minimum rating, shall be installed in the engine block or in the lower coolant inlet hose. The thermostat, with an operating range of 140°F to 160°F, shall be installed.

b. Immersion type engine oil heater, 300 watt minimum rating, shall be installed in oil pan through any convenient opening. The oil heater shall be controlled by its own thermostat not by the coolant heater thermostat.

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c. Battery heater shall maintain battery electrolyte at a temperature of not less than 10°F during vehicle exposure in ambient air temperatures as low as -40°F and shall embody a thermostat to limit the temperature of the electrolyte to not more than 80°F.

d. A fuel warmer or preheater shall be provided to prevent clogging of fuel filters due to wax crystallization in the fuel. The fuel warmer shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of -40°F to an outlet temperature of 9°F, with a flow rate not less than the maximum fuel demand of the engine fuel system.

3.10 Interchangeability. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification to parts on the loader-backhoe.

3.11 Maintainability. Installation of accessories and equipment shall permit ready accessibility for service, replacement and adjustment of component parts with minimum disturbance of other elements.

3.12 Loader. The loader mechanism shall be equipped with the manufacturers general purpose bucket, without teeth, of at least 1.5 cubic yard capacity and shall, as a minimum, have the following capabilities:

a.	Lift capacity to full height	6500 pounds
b.	Height to bucket hinge pin	11 ft.- 0 in.
c.	Clearance with bucket dump at 45 degrees	8 ft.- 8 in.
d.	Minimum breakout force	10,000 pounds

When specified (see 6.2), a 4-in-1 multipurpose bucket minimum capacity of 1.25 cu yd shall be supplied in place of the 1.5 cu yd general purpose bucket.

3.13 Backhoe. The backhoe shall be hydraulically operated and controlled. It shall be a type "C" backhoe in accordance with SAE J326. Controls for the separate and independent operation of the backhoe attachment shall be provided. The backhoe shall have a minimum digging radius of 160° (80° left and 80° right), and shall be equipped with hydrauically controlled stabilizers with street pads. The backhoe shall be equipped with an 18 inch bucket with teeth unless otherwise specified (see 6.2). The backhoe shall contain a mechanism to lock the boom into a safe position for transport and shall have the following minimum capabilities:

a.	Minimum digging depth	15 ft.- 6 in.
b.	Boom swing arc	180 degrees
c.	Minimum bucket digging force	9500 pounds
d.	Boom lift to 12 feet	3000 pounds

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3.13.1 Telescopic Type Extendable Boom. When specified (see 6.2), the standard backhoe boom shall be replaced with a telescopic type extendable boom. When extended the backhoe shall have a minimum digging depth of 19 feet, a minimum lift capacity of 1,800 lbs at a height of 12 feet and a minimum bucket cylinder digging force of 7,500 lbs.

3.14 Engine. The loader-backhoe shall be equipped with a commercial diesel engine, which shall be the current production model of a well established manufacturer. The engine shall demonstrate the performance characteristics specified herein when operating on diesel fuel in accordance with VV-F-800 and Grade JP-5 or JP-8 turbine fuel, the primary fuel, in accordance with MIL-T-83133. Any engine that is adversely affected by either of the above fuels is not acceptable.

3.14.1 Power and speed rating. The power and speed rating of the engine shall be such that operation of the loader-backhoe at specified operating conditions will not require horsepower in excess of the intermittent brake horsepower of the engine for the applicable speed as established by the engine manufacturers performance data. The governed speed and power shall not exceed the highest speed and power for intermittent rated operation at which the engine was previously qualified by the engine manufacturers performance data.

3.14.2 Governing system. The manufacturers current standard governing system will be acceptable, providing it meets the following requirement: Governed engine speed at any load between no-load and full-rated load shall not exceed the full-rated load speed by more than 10 percent.

3.14.3 Engine cooling system. The engine cooling system shall have adequate capacity to prevent overheating under normal operating conditions at the temperature specified in section 3.9 Temperature Conditions.

3.14.4 Air filter. The air filter shall be the manufacturers current commercial dry type.

3.15 Instruments. The loader-backhoe shall be equipped with the following instruments, which shall be installed as specified:

a. Fuel gage or warning lights.

b. Engine lubricating oil pressure gage or warning light.

c. Engine coolant temperature gage or warning light.

d. Ammeter, voltmeter or warning light.

e. Hour meter.

f. Torque converter temperature gage or warning light.

g. Air gage, if air equipped.

Any additional instruments that are standard shall be furnished. Instruments

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may be commercial standard, providing they are weatherproof and dustproof. They shall be of such accuracy and so mounted as to function under all operating conditions specified. All instruments, except the hour meter and fuel gage, shall be located on a panel fastened to the dash in a manner so as to reduce shock and vibration. The hour meter may be mounted in the engine compartment. Fuel gage may be located on the fuel tank. If warning lights are provided, they shall all illuminate when the starter is actuated.

3.16 Transmission. Transmission shall be of the full range power shift type, enabling shifts from any speed range to any other, forward or reverse, with out clutching, with throttle fully depressed. The transmission shall provide a speed of at least 17.5 mph forward and 5 mph reverse. An attachment or control mechanism shall be provided to prevent the engine from starting until the transmission is in neutral.

3.17 Neutral start. An attachment or control mechanism shall be provided to prevent the engine from starting until the transmission is neutralized.

3.18 Hydraulic system. The manufacturers current standard production system which meets or exceeds the specified requirements will be acceptable. The system shall be fully protected from dirt and contamination and shall have sufficient capacity to ensure fast acting control when fully loaded. The maximum working pressure shall be not more that 3000 pounds per square inch (psi). High pressure oil lines shall be of steel tubing as far as practicable and shall meet or exceed the requirements of SAE J524 or SAE J525. When hose is required, the hose assemblies shall meet or exceed the requirements of SAE J517. When specified (see 6.2), auxiliary hydraulic circuits to power front and/or rear mounted attachments shall be provided.

3.18.1 Pump. The pumps shall be either vane or gear type and shall be powered directly by the engine.

3.18.2 Hydraulic cylinder. Each cylinder shall have a stroke of sufficient length to position the components as specified herein.

3.18.3 Piston rods. The exposed surface of the piston rods shall be hard chrome plated in accordance with the manufacturers standard practice.

3.18.4 Reservoir. The reservoir shall be protected against contamination from dirt or other foreign material. An air breather shall be furnished, except when a closed system is provided. A magnetic drain plug or a doughnut-type magnet shall be provided in the reservoir. Design of the reservoir shall include means for checking the fluid.

3.19 Steering. The steering system shall be power or power assisted and shall provide constant velocity steering. Provision shall be made for positive and constant control of the steering at low engine rpm.

3.20 Seat. The loader-backhoe shall be equipped with one or two seats. If a single seat is used, it shall be the turn around type. In either case, the seat(s) shall be well padded with a contoured back support, adjustable fore, aft, up and down for maximum operator comfort.

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3.21 Service brakes. Service brakes shall be hydraulic or air type. Brakes shall be effective on at least two wheels or equivalent, shall be foot operated, and shall stop the unit from a speed of 16 mph within 25 feet on dry, smooth, level concrete surface. Heating of the brakes by repeated stops shall not adversely affect the stopping distance.

3.21.1 Parking brakes. The parking brakes shall be either hand or foot operated and shall be provided with a ratchet or equivalent for locking in place. On a level, brushed, concrete surface the brakes shall hold the rear wheels when a force equal to 35 percent of the weight of the unit is applied to either end.

3.22 Four wheel drive. When specified (see 6.2), the manufacture's standard four wheel drive system shall be provided.

3.23 Differential lock. The manufacture's standard rear differential lock system or a torque proportioning axle with Max-Trac self locking differential shall be provided.

3.24 Tires. Rear tires shall have R-4 type tread. All tires shall be tubeless. Front and rear tires shall be of a size and ply sufficient to carry the rated load at 5 mph speed rating in accordance with the recommendations of the tire and rim association, Incorporated year book. In addition, tires shall also be rated for the G loading as required for Air Transportability (see 3.28).

3.25 Electrical system. The manufacturers current standard commercial system will be acceptable, providing it meets the specified requirements.

3.25.1 Charging system. The charging system shall be the alternator type and shall be of adequate size to maintain the battery(s) in a fully charged state under normal operating conditions.

3.25.2 Battery. The battery(s) shall be the heavy duty type regularly provided for cold weather start.

3.25.3 Horn. The loader-backhoe shall be equipped with an electric horn. The sound level of the horn shall be louder than the sound level of the loader-backhoe operating at full power.

3.25.4 Lighting. The loader-backhoe shall be equipped with a lighting system as follows:

a. Two sealed beam headlights on the front and a lighted instrument panel. The headlights shall be guarded and located to provide forward road illumination when the bucket is raised to travel position.

b. Two sealed beam adjustable flood lights mounted on the rear, one on the left side and one on the right side. The lights shall be located to illuminate the work area to the rear of the loader-backhoe.

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3.26 Fenders. The loader-backhoe shall be equipped with fenders or guards on the rear wheels to prevent mud, water and stones from hitting the operator. When fenders or guards are located so that they may be used as access platforms for servicing or repairs, step pads with non skid abrasive or expanded metal surface shall be provided.

3.27 Identification marking. A corrosion resistant metal nameplate, permanently and legibly marked with the following information, in accordance with MIL-STD-130, to each loader backhoe.

Nomenclature

Manufacturers model number

Manufacturers serial number

Manufacturers name

Date of manufacture

Special characteristics (H.P., RPM, etc)

National Stock Number (NSN)

Contract Number

Curb Weight

Center of gravity (ie., 5 ft. AFT. of FR. Axle)

US

3.28 Lubrication directive. Two charts and one plate shall be furnished with each loader-backhoe. The plate shall be securely attached to each loader-backhoe, and the charts shall be placed in a plastic folder or laminated between clear plastic sheets. The following information shall be furnished on charts and plate:

Point of application

Service interval

Type of lubricant

Viscosity

Federal or Military Specification number (if available)

3.29 Bolted connections. Bolt holes shall be accurately punched or drilled and then deburred. Washers and/or lockwashers shall be provided, all bolts, nuts and screws shall be securely tightened in accordance with good commercial practice.

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3.30 Welding. The surfaces of all parts to be welded shall be free from rust, scale, paint, grease or other foreign matter. Spot, tack or intermittent welds for strength will not be permitted. Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Fillet weld shall be provided when necessary to reduce stress concentration. Welding shall be in accordance with a nationally recognized welding code.

3.31 Air transportability. The vehicle shall be air transportable in accordance with MIL-STD-1791 and AFSC DH1-11 in types C130, C141 and C5 aircraft. Removal of top mounted equipment shall be permitted in order to meet height requirements. Removal and reinstallation of equipment shall not require more than 3 man hours for each operation. Removal of equipment shall in no way prevent loading/unloading of the unit under its own power. Equipment removal shall only require the use of common hand tools and a lifting unit, such as a fork lift or front end loader. Address any questions concerning air transportability to ASC/ENECA, Wright Patterson AFB OH 45433, phone number 513-255-6039.

3.32 Lifting and Tie Down Attachments. When specified (see 6.2), the equipment shall be equipped with lifting and tying down attachments. Lifting and tie down attachments shall conform to Type II or Type III of MIL-STD-209. A non-ferrous transportation plate shall be transcribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tying down attachments may be identified by stencilling or other suitable marking. Tie-down markings shall clearly indicate that the attachments are intended for the tie down of the equipment on the carrier when shipped.

3.33. Maintainability. All maintenance described in this specification shall be capable of being performed by one mechanic using common hand tools. Design and construction shall permit ready accessibility to all parts for maintenance and repair in the field. Replacement and adjustment of component parts and accessories shall be accomplished with minimum drainage of fluids and disturbance to other parts.

3.34 Filling, draining and checking provisions. All fluid systems shall possess a convenient means of filling, draining and checking levels of the system. All drains shall conduct fluids directly to the ground in a stream without onto other components. Fluid level indicators or fill ports shall identify the type of fluid, state whether the level should be checked hot or cold and indicate "full" and "add".

3.35 Maintenance operations list. Any four or more operations shall be able to be performed by one person in the sum of times allotted for each operation.

a. Remove, replace and adjust all engine driven belts - 1 hour.

b. Remove and replace voltage or current regulator - 15 minutes.

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c. Remove and replace generator and adjust generator belt - 30 minutes.

d. Remove and replace all filter elements, screens and strainers on the hydraulic system - 30 minutes.

e. Drain cooling system, remove and replace engine cooling hoses and refill cooling system - 1 hour.

f. Drain engine lubricating oil, remove and replace oil filter element and refill crankcase - 30 minutes.

g. Remove all replaceable fuel filter elements - 30 minutes.

h. Remove and replace transmission fluid filter elements and strainer - 45 minutes.

i. Drain and refill transmission and torque converter - 20 minutes.

j. Disconnect battery leads, remove and replace batteries and reconnect battery leads - 15 minutes.

3.36 Workmanship. Workmanship of the loader-backhoe, related equipment and accessories shall be of a quality consistent with the Industries normally produced equipment of the type specified.

3.37 Rustproofing. When specified (see 6.2), the following areas of the backhoe-loader shall be treated for rustproofing:

(1) Hood
(2) Roof
(3) Doors
(4) Pillars
(5) Cab interior
(6) Other body requirements (cab only)
(7) See FED-STD-297, para 5.8 for areas of tropical rustproofing

NOTE: When rustproofing is specified, first article test units shall not be rustproofed until approval of the FAT report.

4. Testing.

4.1 Responsibility for testing. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications, where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as

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follows:

a. First Article Inspection.

b. Quality conformance Inspection.

4.2.1 First article inspection. First article inspection shall be applied to the first article sample submitted in accordance with 3.1. Failure of the preproduction sample to pass the examination of one or more first article tests shall be cause for rejection. First article inspection shall consist of the examination in 4.3 and all tests in 4.4. These tests shall be performed using JP-8, JET A or Jet A-1.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Failure of the machine to pass the examination of one or more first article tests shall be cause for rejection. Quality conformance shall consist of the following requirements:

a. Examination (4.3).

b. Test (4.4.1).

c. Inspection of preservation, packaging, packing and marking for shipment and storage (4.5).

4.3 Examination. The loader-backhoe and equipment shall be examined for design, construction, materials, components, electrical equipment and workmanship to determine compliance with the requirements of this specification. Any deviation from these requirements shall be cause for rejection.

4.4 Tests.

4.4.1 Operational test. Each loader-backhoe shall be completely assembled, adjusted, lubricated and otherwise serviced for operation. Engine shall be started and after a warm-up period, a run-in test shall be conducted in accordance with the manufacturers recommendations. During the operational test, all controls, adjusting mechanisms and other accessories shall be operated sufficiently to ensure proper functioning against restrictions or possible malfunction. Failure to pass any phase of this test shall be cause for rejection.

4.4.2 Speed test. The loader-backhoe shall be tested to determine the forward and backward speed range. Failure to comply with 3.16 shall be cause for rejection.

4.4.3 Environmental test. The supplier shall furnish a certification that his equipment will perform satisfactorily at the specified temperatures and will not be damaged from storage under the specified temperatures cited in 3.9.

4.4.4 Lifting and tie-down attachments testing. When lifting and tie-down

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attachments are required, the lifting and tie-down attachments shall be tested to determine compliance with 3.32.

4.4.5 Performance test. The loader-backhoe shall be tested to determine its ability to meet the requirements of 3.12 and 3.13. Failure to pass any phase of this test shall be cause for rejection. When telescopic type extendable boom is required, it shall be tested to determine compliance with 3.13.1.

4.4.6 Brake test. The loader-backhoe shall be tested to determine that the brakes meet the requirements of 3.21 and 3.21.1. Failure to meet these requirements shall be cause for rejection.

4.4.7 Maintenance testing. After having performed the inspection according to 4.3, an authorized government representative shall select between four and all of the above tasks listed in 3.35. These tasks shall be performed sequentially by one person in the sum of time allotted for each task. All tools, parts and supplies shall be on hand at the start of the test. This list of tasks performed and the total time taken to perform the tasks shall be recorded. The use of any tools other than common hand tools shall be noted. Power tools shall not be used.

4.4.8 Air transportability test. The manufacturer shall demonstrate that the loader-backhoe meets all transportability requirements specified in 3.31. The necessary information to demonstrate air transportability as required, shall be submitted to ASC/ENECA, Wright Patterson AFB OH 45433 (Phone 513-255-6039), for approval. This information shall also be included in the first article test report.

4.4.9 Sound level testing.

4.4.9.1 Exterior sound level test. The exterior of the vehicle shall be tested in accordance with MIL-STD-1474b to demonstrate compliance with 3.5.1.1.

4.4.9.2 Cab interior sound level test. When cab is required, the cab interior sound level shall be tested in accordance with MIL-STD-1474b to demonstrate compliance with 3.5.1.2.

4.4.10 Certifications required. The contractor shall certify compliance with the following:

29 CFR 1926.600

29 CFR 1926.602

29 CFR 1926.1000

29 CFR 1926.1003

4.5 Inspection of preservation, packaging, packing and marking for shipment and storage. Preservation, packaging, packing and marking for shipment and storage shall be inspected to verify conformance to the requirements of

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Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The preservation and packaging shall be level A or C, as specified in 6.2.

5.1.1 Level A. The complete loader-backhoe shall be preserved and packaged in accordance with the level A, type III, unboxed (mobile requirements of Specification MIL-T-3351).

5.1.2 Level C. The complete loader-backhoe shall be preserved and packaged in accordance with the suppliers standard practice.

5.2 Packing. The packing shall be Level A, B or C as specified in 6.2.

5.2.1 Level A. The complete loader-backhoe shall be packed in accordance with the Level A, Type III, unboxed (mobile requirements of Specification MIL-T-3351).

5.2.2 Level B. The complete loader-backhoe shall be packed in accordance with the level B, Type III, unboxed (mobile requirements of Specification MIL-T-3351).

5.2.3 Level C. The complete loader-backhoe shall be packed or prepared for shipment in a manner to ensure undamaged and unmarred arrival at destination and which will be acceptable to carriers at their lowest rates.

5.3 Marking. The loader attachments, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTE

6.1 Intended use. The loader-backhoe is intended primarily for use in bucket loader operations, long-range stock pile work, excavating and as a replacement for small shovels.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. First article sample, if required (see 3.1).

c. Sealed cab, if required (see 3.2.1).

d. A personnel heater, defroster and/or defroster fan or air conditioner or any combination thereof, if required (see 3.2.1).

e. Temperature condition, if different than specified (see 3.9).

f. A miltipurpose or general purpose bucket, if required (see 3.12).

g. Type bucket required for backhoe, if other than specified (see 3.13).

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h. Telescopic type extendable boom, if required (see 3.13.1).

i. Auxiliary hydraulic circuits to front and/or rear of backhoe-loader, if required (see 3.18).

j. Four wheel drive system, if required (see 3.22).

k. Lifting and tie down attachments, if required (see 3.32).

l. Rustproofing, if required (see 3.37).

m. Any additional requirements shall be as specified.

n. Level of preservation, packaging and level of packing required (see 5.1 and 5.2).

6.3 First article sample. The first article sample may be a production item which meets the requirements of this specification. The sample shall be made using the same methods used in production of the items to fulfill the terms of the contract. Approval of a first article sample shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of this specification. The first article sample may be shipped as part of the contract.

6.4 Subject term (key word) listing.

a. Loader.

b. Backhoe.

c. Excavator.

d. Bucket, Scoop.

e. Pneumatic Tired.

f. Engine, Diesel.

Custodian:
Air Force - 99
Army - AT
Navy - YD

Preparing Activity:
Air Force - 84

Project Number:
3805-0155

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